

Calculus Solutions

1. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + 2xy + 3y^2}{3x^2 + 2xy + y^2}$ does not exist.

Solution. Let $f(x, y) = \frac{x^2 + 2xy + 3y^2}{3x^2 + 2xy + y^2}$ for $(x, y) \neq (0, 0)$. Then $f(x, 0) = 1/3$ for all $x \neq 0$, and $f(0, y) = 3$ for all $y \neq 0$. So there are points arbitrarily close to $(0, 0)$ where $f(x, y) = 1/3$, and others where $f(x, y) = 3$. Therefore the limit doesn't exist.

2. Find $\frac{\partial z}{\partial x}$ if $xy \cos z + y^3 z \sin x = 17$.

Solution. We differentiate both sides of the equation with respect to x , treating y as a constant, and z as a function of x , using the product rule and chain rule:

$$\begin{aligned}\frac{\partial}{\partial x}(xy \cos z + y^3 z \sin x) &= \frac{\partial}{\partial x} 17 \\ y \frac{\partial}{\partial x}(x \cos z) + y^3 \frac{\partial}{\partial x}(z \sin x) &= 0 \\ y(\cos z - x \sin z \frac{\partial z}{\partial x}) + y^3(z \cos x + \sin x \frac{\partial z}{\partial x}) &= 0\end{aligned}$$

and now solve for $\frac{\partial z}{\partial x}$;

$$\frac{\partial z}{\partial x} = \frac{y \cos z + y^3 z \cos x}{xy \sin z - y^3 \sin x}$$

which can be simplified to

$$\frac{\partial z}{\partial x} = \frac{\cos z + y^2 z \cos x}{x \sin z - y^2 \sin x}$$

3. Find the derivative of $f(x, y) = 2xy + 3x + 4y + 5$ at $(1, 2)$ in the direction $(2, 3)$.

Solution. The unit vector in the direction $(2, 3)$ is $(2/\sqrt{13}, 3/\sqrt{13})$. The formula for the directional derivative is

$$\left(\frac{\partial f}{\partial x}(1, 2), \frac{\partial f}{\partial y}(1, 2) \right) \cdot \left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right)$$

We compute $\frac{\partial f}{\partial x} = 2y + 3$, $\frac{\partial f}{\partial y} = 2x + 4$, so $\frac{\partial f}{\partial x}(1, 2) = 7$ and $\frac{\partial f}{\partial y}(1, 2) = 6$.

The answer is $32/\sqrt{13}$.

Calculus Solutions

1. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + 2xy + 3y^2}{x^2 - 2xy + 3y^2}$ does not exist.

Solution. Let $f(x, y) = \frac{x^2 + 2xy + 3y^2}{x^2 - 2xy + 3y^2}$ for $(x, y) \neq (0, 0)$. Then $f(x, 0) = 1$ for all $x \neq 0$, and $f(x, x) = 3$ for all $x \neq 0$. So there are points arbitrarily close to $(0, 0)$ where $f(x, y) = 1$, and others where $f(x, y) = 3$. Therefore the limit doesn't exist.

2. Find $\frac{\partial z}{\partial x}$ if $xy \log z + y^3 z \log x = 17$.

Solution. We differentiate both sides of the equation with respect to x , treating y as a constant, and z as a function of x , using the product rule and chain rule:

$$\begin{aligned} \frac{\partial}{\partial x}(xy \log z + y^3 z \log x) &= \frac{\partial}{\partial x} 17 \\ y \frac{\partial}{\partial x}(x \log z) + y^3 \frac{\partial}{\partial x}(z \log x) &= 0 \\ y(\log z + \frac{x}{z} \frac{\partial z}{\partial x}) + y^3(\frac{z}{x} + \log x \frac{\partial z}{\partial x}) &= 0 \end{aligned}$$

and now solve for $\frac{\partial z}{\partial x}$;

$$\frac{\partial z}{\partial x} = -\frac{y \log z + y^3 z x^{-1}}{y x z^{-1} + y^3 \log x}$$

which can be simplified to

$$\frac{\partial z}{\partial x} = -\frac{xz \log z + y^2 z^2}{x^2 + y^2 x z \log x}$$

3. Find the derivative of $f(x, y) = 5xy + 4x + 3y + 2$ at $(2, 1)$ in the direction $(3, 2)$.

Solution. The unit vector in the direction $(3, 2)$ is $(3/\sqrt{13}, 2/\sqrt{13})$. The formula for the directional derivative is

$$\left(\frac{\partial f}{\partial x}(2, 1), \frac{\partial f}{\partial y}(2, 1) \right) \cdot \left(\frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}} \right)$$

We compute $\frac{\partial f}{\partial x} = 5y + 4$, $\frac{\partial f}{\partial y} = 5x + 3$, so $\frac{\partial f}{\partial x}(2, 1) = 9$ and $\frac{\partial f}{\partial y}(2, 1) = 13$.

The answer is $53/\sqrt{13}$.

Algebra

4. Put $A = \begin{bmatrix} 2 & 2 & 3 & 4 \\ 0 & 2 & 3 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

- (a) Write down the eigenvalues of the matrix A .
- (b) Find a basis for the eigenspaces $E_\lambda(A)$ of each of the eigenvalues λ .
- (c) Is A diagonalizable? Give reasons.

Solution (a) By inspection (since the matrix is triangular), the eigenvalues are 1, 1, 2, 2.

(b) $\lambda = 1$: The eigenspace consists of the solutions to the linear system

$$2w + 2x + 3y + 4z = w$$

$$0w + 2x + 3y + 2z = x$$

$$0w + 0x + 1y + 1z = y$$

$$0w + 0x + 0y + 1z = z$$

That is, $w = -x$, $x = x$, $y = -\frac{1}{3}x$, $z = 0$. Any non-zero choice of the free variable x will determine a basis for the eigenspace; for example, $(w, x, y, z) = (3, -3, 1, 0)$.

$\lambda = 2$: The eigenspace consists of the solutions to the linear system

$$2w + 2x + 3y + 4z = 2w$$

$$0w + 2x + 3y + 2z = 2x$$

$$0w + 0x + 1y + 1z = 2y$$

$$0w + 0x + 0y + 1z = 2z$$

That is, $x = y = z = 0$. Any non-zero choice of the free variable w will determine a basis for the eigenspace; for example, $(w, x, y, z) = (1, 0, 0, 0)$.

(c) The matrix A is not diagonalizable since there are at most two linearly independent eigenvectors; there would need to be 4 for diagonalizability.

Algebra

4. Put $A = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$.

- (a) Write down the eigenvalues of the matrix A .
 (b) Find a basis for the eigenspaces $E_\lambda(A)$ of each of the eigenvalues λ .
 (c) Is A diagonalizable? Give reasons.

Solution (a) By inspection (since the matrix is triangular), the eigenvalues are 1, 1, 3, 3.

(b) $\lambda = 1$: The eigenspace consists of the solutions to the linear system

$$1w + 1x + 1y + 2z = w$$

$$0w + 1x + 2y + 4z = x$$

$$0w + 0x + 3y + 0z = y$$

$$0w + 0x + 0y + 3z = z$$

That is, $x = y = z = 0$. Any non-zero choice of the free variable w will determine a basis for the eigenspace; for example, $(w, x, y, z) = (1, 0, 0, 0)$.

$\lambda = 3$: The eigenspace consists of the solutions to the linear system

$$1w + 1x + 1y + 2z = 3w$$

$$0w + 1x + 2y + 4z = 3x$$

$$0w + 0x + 3y + 0z = 3y$$

$$0w + 0x + 0y + 3z = 3z$$

That is, $w = x$, $x = x$, $y = x - 2z$, $z = z$. Any non-trivial choice of the free variables x and z will determine a basis for the eigenspace; for example, $(1, 1, 1, 0)$ and $(0, 0, -2, 1)$.

(c) The matrix A is not diagonalizable since there are at most three linearly independent eigenvectors; there would need to be 4 for diagonalizability.